Connecting the Dots in AMR Surveillance

How bacteria climb the resistance ladder, and how to forecast the next rung

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Abstract

Antibiotic resistance arises through mechanisms such as drug inactivation, target modification, reduced permeability and active efflux. Because one mechanism can affect several drugs and distinct resistance determinants can accumulate or travel together, resistance profiles need not be arbitrary [1,2]. Drugs may instead form a resistance ladder: those defeated with a narrower repertoire of capabilities lie on lower rungs, while those requiring a broader repertoire lie higher. Routine surveillance, read one pathogen–antibiotic pair at a time, cannot reveal this structure. We recover it from the Pfizer ATLAS programme distributed through the Vivli AMR Register, comprising 394 species, 43 antibiotics and observations from 2004–2024 [3]. Rather than considering absolute resistance alone, we ask whether each species is more resistant to each drug than expected from the species' overall resistance and the drug's overall failure rate. Linking species to the drugs they comparatively defeat yields a bipartite network that is strongly nested within coherent organism–drug groups: narrow comparative-resistance profiles are largely contained within broader ones. Fitness–Complexity converts this ladder into coupled rankings of species and drugs [4]. Without using clinical labels, drug complexity rises across the WHO Access, Watch and Reserve tiers, while ESKAPEE taxa occupy the upper rungs of their local ladders [5,6]. The ladder also anticipates the next rung. Connecting antibiotics defeated by the same species defines an empirical similarity network; a parameter-free exposure score reached an AUC of 0.71 for the following year and remained informative over five years. The framework is delivered as an interactive atlas (`https://giordano-demarzo.github.io/resistome-atlas/`) showing, for any pathogen, where it stands and which drugs it is closest to defeating, with source code and reproducibility materials available on GitHub [7].

Objectives

We pursued three linked objectives: (*i*) turn routine counts into a pathogen–antibiotic network that isolates pair-specific comparative resistance from the general resistance level of a species and the general vulnerability of a drug; (*ii*) test whether comparative-resistance profiles are nested, as partially ordered accumulation of mechanisms predicts [8,9], and use Fitness–Complexity to place every species and drug on the resulting ladder; and (*iii*) test prospectively whether a species' position relative to empirically similar drugs anticipates which drug it defeats next.

Methods

From surveillance counts to a pathogen–antibiotic network

The input contains annual species–antibiotic counts from the Pfizer ATLAS surveillance programme distributed through the Vivli AMR Register, spanning 394 species, 43 antibiotics and 2004–2024 [3]. For each analysis window, resistant and tested counts were pooled before applying the minimum-isolate threshold. A pair (p, a) was retained when at least $n_{\min} = 30$ isolates had been tested, and its resistance fraction was $r_{pa} = n^R_{pa}/N_{pa}$, where intermediate isolates contribute to N_{pa} but not to n^R_{pa}. If the pair was absent or below threshold, r_{pa} was undefined. The workflow was repeated at $n_{\min} = 10$ and 50, all else unchanged.

Raw percentages alone do not isolate pair-specific structure. A resistance fraction of 40% may reflect a broadly resistant species, a drug that fails broadly, or a species–drug pair that is unusually resistant after accounting for both margins. Only the third is the comparative signal of interest: it may be compatible with enrichment or acquisition of a resistance mechanism, but RCA alone cannot identify one. We therefore divide the observed fraction by the fraction expected from the two margins. With $\mathcal{O}$ the sufficiently observed cells, $R_p = \sum_{a:(p,a)\in\mathcal{O}} r_{pa}$ the overall resistance level of species p, $A_a = \sum_{p:(p,a)\in\mathcal{O}} r_{pa}$ the overall failure level of drug a, and $T = \sum_{(p,a)\in\mathcal{O}} r_{pa}$,

$$\mathrm{RCA}_{pa} = \frac{r_{pa}}{R_p A_a / T} = \frac{r_{pa} T}{R_p A_a}$$

so $\mathrm{RCA}_{pa} = 2$ means twice the resistance implied by the two margins. This is the revealed comparative advantage construction, introduced to compare economic profiles of unequal size and used here for the same reason [10]. It is invariant to uniform escalation: if resistance rises proportionally everywhere, every RCA is unchanged. The analysis therefore tracks pair-specific comparative resistance rather than the general trend already reported by prevalence. A high value can also occur at a modest absolute percentage, potentially highlighting a pair before high prevalence is reached.

We call $\mathrm{RCA}_{pa} \geq 1$ a *comparative-resistance link* and write $M_{pa} = \mathbf{1}\{\mathrm{RCA}_{pa} \geq 1\}$. The binary matrix M defines a bipartite comparative-resistance network, with species on one side and antibiotics on the other; in the pooled 2015–2024 window it contains 139 species, 40 drugs and 991 links. Everything that follows is computed from this object. Undefined RCA evaluates as false, so $M_{pa} = 0$ merges observed pairs with $\mathrm{RCA}_{pa} < 1$ and insufficiently observed pairs: zero means "no comparative-resistance link in the analysed universe", not confirmed susceptibility. RCA is a within-dataset comparative signal, not a clinical breakpoint.

Nested profiles: the resistance ladder

If resistance mechanisms accumulate along partially constrained trajectories, the binary comparative-resistance matrix M should have a characteristic shape [8,9]. Species with fewer accumulated capabilities should have fewer links, largely contained within the broader comparative-resistance profiles of species with more capabilities; rarely linked drugs should therefore be reached mainly from the top. Ordering rows and columns by profile breadth would make M approximately triangular. This property is *nestedness*, common in ecological bipartite networks where the partners of specialists form subsets of those of generalists [11,12]. Here nestedness is the structural signature we call the resistance ladder: comparative-resistance profiles form successive rungs of a broadly ordered accumulation process rather than arbitrary drug sets. Because each analysis window is observational, this structure is consistent with–but does not establish–a temporal sequence or its underlying mechanisms.

Nestedness is interpretable only among species facing broadly comparable antibiotic spectra. Intrinsic permeability barriers, taxon-specific mechanisms and testing practices make many organism–drug combinations fundamentally different; for example, the Gram-negative outer membrane restricts entry of many large antibiotics, and susceptibility expert rules differ across taxa [13,14]. Gram-positive and Gram-negative organisms should therefore not automatically be treated as rungs of the same ladder. We partitioned M by spectral co-clustering, assigning species and drugs simultaneously to five matched groups [15]. The partition used no taxonomy, drug class, AWaRe or ESKAPEE labels; its purpose was to define comparable groups within which nestedness could be measured.

For species i, j with degrees k_i, k_j, let $O_{ij} = \sum_a M_{ia} M_{ja}$ be the number of drugs both defeat, with degree-based expectation $\langle O_{ij} \rangle = k_i k_j / A$ over A drug columns. The statistic compares the overlap of a narrower profile with a broader one against this expectation, and analogously for drugs [16,17]:

$$\mathcal{I}_R = \sum_{\substack{i,j \\ k_i > k_j,\, b_i = b_j}} \frac{O_{ij} - k_i k_j / A}{k_j (C_i - 1)} \quad , \qquad \mathcal{I}_C = \sum_{\substack{a,c \\ k_a > k_c,\, b_a = b_c}} \frac{O_{ac} - k_a k_c / P}{k_c (C_a - 1)} \quad , \qquad \mathcal{I} = \frac{2}{P + A} (\mathcal{I}_R + \mathcal{I}_C)$$

where b identifies a group and C_i or C_a its size on the corresponding side of the bipartite network. Setting all nodes to one group gives global nestedness $\mathcal{N}$, measuring how well one ladder describes the entire matrix; the same local–global distinction appears in economic networks [17]. Significance was assessed with 100 Bernoulli matrices satisfying $\Pr(M^{\text{null}}_{pa} = 1) = \min(k_p k_a / E, 1)$, $E = \sum_{pa} M_{pa}$, which preserve degree heterogeneity in expectation. Empty nodes were removed and every null matrix was re-clustered, allowing group structure to arise by chance.

Fitness and complexity: what a species has accumulated, what a drug requires

Fitness–Complexity scores a species and a drug in one coupled calculation, since the two are defined through each other [4]. A species' *fitness* rises with the number of drugs it defeats, each weighted by the drug's *complexity*: defeating a drug that nearly everything defeats says little, defeating one that almost nothing defeats says a great deal. Complexity, in turn, is the accumulated capability a drug requires before it can be defeated, and the calculation infers it from who has managed: complexity is high only if *every* species defeating the drug is well equipped, because the update is harmonic in fitness and a single poorly equipped conqueror collapses the score. This asymmetry is deliberate, since one modestly equipped species defeating a drug is proof that the drug did not require much, and it separates complexity from mere rarity of resistance. Starting from $F^{(0)}_p = Q^{(0)}_a = 1$, the pipeline iterates

$$\widetilde{F}^{(n)}_p = \sum_a M_{pa} Q^{(n-1)}_a \quad , \qquad \widetilde{Q}^{(n)}_a = \left(\sum_p \frac{M_{pa}}{F^{(n-1)}_p} \right)^{-1}$$

normalising F and Q by their means at each step. Because heterogeneous non-nested structures can produce collapsed rankings, scores were computed separately within each group and standardised before pooling. They therefore describe position on a local ladder, not an absolute scale shared across groups. Fitness is a network score, unrelated to evolutionary fitness; complexity is not drug potency or clinical efficacy.

Two external classifications were used only after scoring: the WHO Access, Watch and Reserve stewardship tiers [5] for antibiotic complexity, and the ESKAPEE group, which extends the ESKAPE mnemonic to include *Escherichia coli* [6], for pathogen fitness. Neither entered the network, grouping or scores.

The antibiotic similarity network and prospective forecasting

If two antibiotics are repeatedly defeated by the same species, they have similar comparative-resistance profiles. This co-resistance may reflect shared mechanisms, linked determinants, correlated selection or lineage structure, but phenotypic co-occurrence alone does not identify the cause [2,18]. Projecting the bipartite network onto its antibiotic side makes the empirical similarity explicit: $W^A_{ab} = \sum_p M^{(t)}_{pa} M^{(t)}_{pb}$, giving a weighted similarity network over the 40 antibiotics. A species that already defeats many of a drug's neighbours has high network exposure to that drug; this is an empirical proximity to the next rung, not direct evidence of a shared mechanism. For each origin year $t = 2019, \ldots, 2023$, all quantities were rebuilt using only data from 2004 through t. For every sufficiently observed pair (p, a) with no link at t, exposure was the weighted share of drug a's neighbourhood already defeated by species p:

$$\rho^A_{pa} = \frac{\sum_o W^A_{ao} M^{(t)}_{po}}{\sum_o W^A_{ao}}$$

The outcome was whether the same sufficiently tested pair had $\mathrm{RCA}_{pa} \geq 1$ in year $t + 1$. The score contains no fitted coefficient, training step or tuned threshold. Discrimination was measured by AUC against two baselines: the drug's current fraction of linked species, and the candidate pair's current RCA_{pa} (its proximity to the threshold). Significance used 200 node-label permutations. To test persistence, the 2019 network, scores and candidate set were frozen and evaluated unchanged against 2020–2024, restricting each evaluation to pairs sufficiently tested in that target year.

Results

Comparative-resistance profiles are nested: each group is a ladder

Pooling 2015–2024 yielded 1,932 sufficiently observed cells in a 139-by-40 network, of which 991 carried a comparative-resistance link. Within the five groups, nestedness was pronounced: $\mathcal{I}^* = 0.3855$ against 0.1453 ± 0.0229 for degree-matched randomisations ($z = 10.51$). Narrow comparative-resistance profiles sat inside broader ones, and drugs at the top of each group were linked only to species that defeated almost everything else in it. The groups therefore formed local resistance ladders consistent with partially ordered accumulation [8,9] (Figure 1a).

Applied to M as a whole, the same statistic was low and below the null ($\mathcal{N} = 0.0496$ against 0.0844 ± 0.0081, $z = -4.30$); $\mathcal{I}^*/\mathcal{N} = 7.77$ (Figure 1b). This is consistent with combining organisms that differ in intrinsic barriers, resistance repertoires and testing spectra [13,14]: a *Klebsiella* and an *Enterococcus* do not lie on one biologically interpretable axis. As a qualitative check, the five data-derived groups matched recognisable organism–drug domains: one large Gram-negative group; separate staphylococcal and enterococcal groups; and two anaerobe groups centred on *Clostridia* and *Bacteroides*, each paired with the drugs represented in its testing profile.

Fitness and complexity place species and drugs on the ladder

Within each group the equations converged to a stable fitness for every species and complexity for every drug. Both rankings agreed with labels withheld from the construction. Antibiotic complexity increased across WHO AWaRe tiers (Spearman $\rho = 0.5895$, $p = 9.84 \times 10^{-5}$, 38 drugs; $\rho = 0.6244$, $p = 8.49 \times 10^{-4}$ in the largest group), aligning the network ranking with the stewardship gradient [5] (Figure 2a). ESKAPEE taxa [6] occupied the upper rungs: median standardised fitness was 1.044 for 16 matching taxa and -0.387 for 107 others (one-sided Mann–Whitney $p = 4.60 \times 10^{-6}$; Figure 2b). These associations provide external consistency, not clinical validation: both scores use only the topology of comparative-resistance links.

Comparative-resistance profiles expand toward similar drugs, and the next rung can be anticipated

Projecting the bipartite network onto its antibiotic side produced the antibiotic similarity network (Figure 3a), where edge weight counts species linked to both drugs. For *Klebsiella pneumoniae*, comparative-resistance links expanded locally from 12 drugs in 2014 to 17 in 2019 and 21 in 2024, while distant regions remained unlinked (Figure 3b–d). This local expansion is the ladder seen in motion at the level of comparative-resistance profiles and motivates the neighbourhood forecast; it does not by itself establish molecular acquisition or transmission.

The rolling-origin back-test contained 1,839 eligible pathogen–drug pair-years, of which 131 (7.12%) crossed the comparative-resistance threshold in the following year. The exposure ranking clearly separated absolute event rates: 51 of 307 pairs (16.61%) in the top exposure sixth crossed the threshold within one year, compared with 14 of 307 (4.56%) in the bottom sixth, a 3.6-fold difference (Figure 4a). Network exposure, rebuilt at each origin from raw co-occurrence weights in M, reached a pooled AUC of 0.7076 (origin-specific 0.6688–0.7703), compared with 0.5123 for drug prevalence (Figure 4b). An AUC of 0.7076 means that a future event outranked a non-event 70.76% of the time; it measures ranking performance, not individual probability. Exposure also exceeded all 200 node-label permutations ($p < 0.005$). Per-drug AUC was highest for ceftazidime, gentamicin, meropenem, cefepime and ceftriaxone, reaching 0.89 for ceftazidime (Figure 4c). We call these drugs "contagion hubs": a descriptive network label for neighbourhoods that best announce the next comparative-resistance event. Each estimate rests on 5–9 events and remains exploratory. Finally, current RCA proximity reached AUC 0.6835 and was only weakly correlated with exposure (Pearson $r = 0.38$), indicating that the neighbourhood adds information beyond being already close to $\mathrm{RCA} = 1$ (Figure 4d).

The signal persisted beyond one year. With network, scores and candidate set frozen at 2019, AUC remained 0.681–0.742 across one- to five-year horizons, while the prevalence baseline stayed near chance (0.41–0.58). Results were also robust to the isolate threshold: repeating the pipeline at $n_{\min} = 10, 30, 50$ preserved in-group nestedness and ESKAPEE separation. The default $n_{\min} = 30$ gave the strongest AWaRe association ($\rho = 0.590$, versus 0.547 and 0.442) and highest back-test AUC (0.708, versus 0.638 and 0.702).

Impact of the work

What this adds to surveillance practice. A surveillance report answers "how much resistance is there?", not "where is it going?". From the same counts, this framework adds three views: a *position* for every species and drug on its local ladder; an antibiotic *similarity network* built from comparative-resistance behaviour rather than chemical class; and a *ranked watch-list* of pairs that lie near existing comparative-resistance links. Contagion hubs also nominate drugs for prospective evaluation as early-warning sentinels when testing capacity is limited.

Interpretability is a design requirement, not a claim. The chain is short and inspectable (counts $\to r_{pa} \to \mathrm{RCA}_{pa} \to M \to$ similarity network $\to$ exposure score): no classifier is trained, coefficient fitted or threshold tuned. A pair ranks highly because specific drugs in its neighbourhood are already linked to that species, and this rationale can be checked against the isolate counts and challenged by a microbiologist. The analysis can be recomputed as soon as new counts arrive.

The Resistome Atlas. The interactive dashboard (`https://giordano-demarzo.github.io/resistome-atlas/`) is delivered with a deterministic end-to-end script and output tables. Users can follow a pathogen's comparative-resistance profile through the antibiotic similarity network, inspect local ladder positions alongside AWaRe labels, and view the drugs it is closest to defeating. Surveillance teams can identify climbing organisms, stewardship groups can inspect restricted drugs near existing links, and laboratories can prioritise candidate pairs for confirmatory testing. Dashboard probabilities come from a separate one-variable calibration of exposure, not from the parameter-free evaluation reported here; no output is clinical guidance.

Deployment beyond this dataset. The pipeline needs only isolate counts by species, drug and year. Because comparative resistance is defined relative to the analysed population, fractions, margins, groups and networks must be recomputed for each country, region or time window; one hospital network need not share the global ladder. Temporal re-estimation is demonstrated here, whereas geographical re-estimation remains a deployment opportunity because the present table pools countries. Genomic integration could test ladder-implied relationships against known resistance genes and mutations, while recognising that genotype–phenotype correspondence is incomplete [19].

Limits. Comparative resistance is relative: a pair may be flagged at modest prevalence, or remain unremarkable despite clinically severe absolute resistance. ATLAS is a surveillance sample rather than a population-random sample, so representativeness may be limited, as in laboratory-based surveillance more generally [20]. A zero in M also merges observed pairs below the threshold with insufficiently tested pairs, so the network partly reflects testing coverage. The number of groups is fixed at five, the null preserves degree only in expectation, and AWaRe and ESKAPEE provide external consistency rather than clinical validation. AUC measures ranking, not individual risk; changes in comparative status may reflect shifting margins or sampling composition as well as biology; and nothing here establishes causation, transmission or treatment failure.

Tables/figures

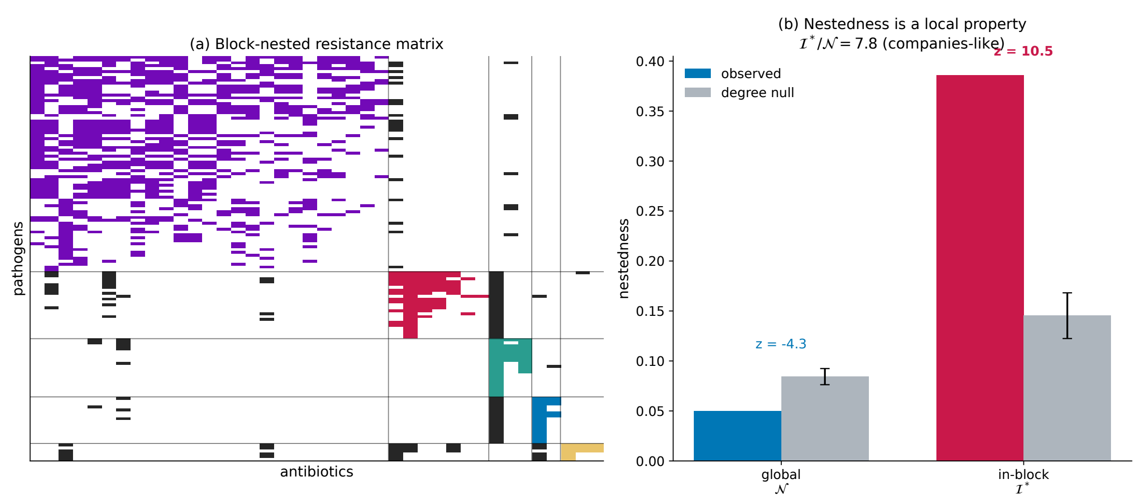

Figure 1: Comparative-resistance profiles are nested within groups. (a) Species and drugs are ordered by the five data-derived groups and by profile breadth within each group. Coloured cells are comparative-resistance links inside matched groups; dark cells connect different groups. Local blocks show the triangular pattern of the resistance ladder: narrow profiles sit inside broader ones. (b) Global nestedness $\mathcal{N}$ and in-group nestedness $\mathcal{I}^*$ against the degree-matched Bernoulli null. One ladder spanning all organisms performs below chance; local ladders perform far above it.

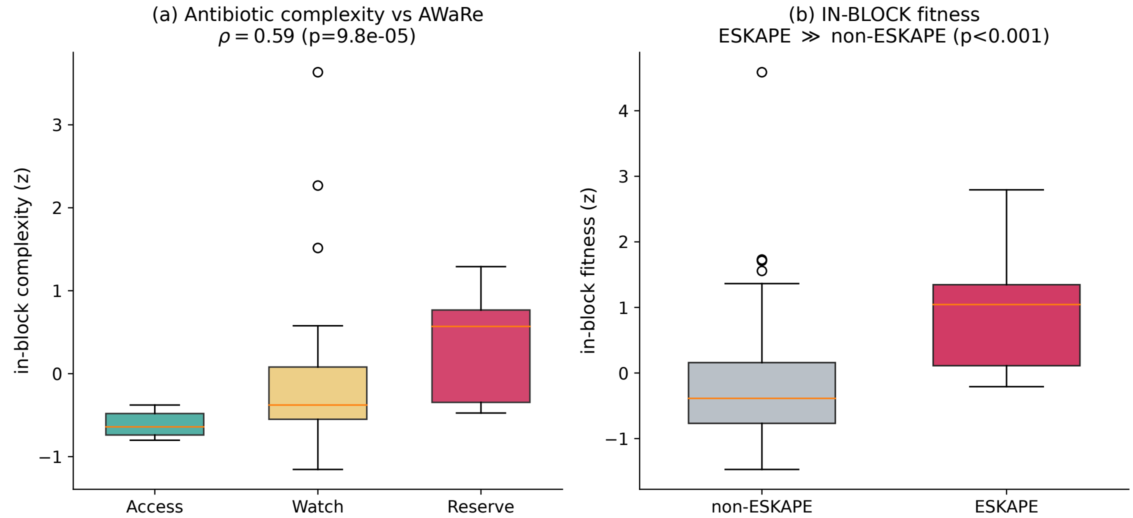

Figure 2: Ladder position recovers withheld biology. (a) Within-group drug complexity rises across WHO AWaRe tiers [5]. (b) ESKAPEE taxa [6] occupy the upper rungs of their local ladders. Neither label was used to build the network, define groups or compute scores. Fitness is a network score, not evolutionary fitness; complexity is not drug potency.

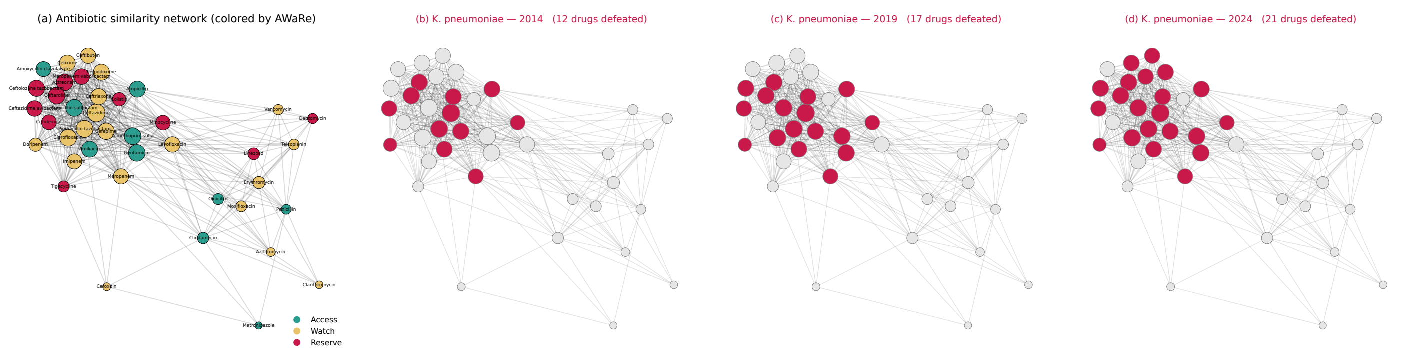

Figure 3: The antibiotic similarity network and expansion of comparative-resistance profiles. (a) Drugs are joined in proportion to the number of species linked to both; colour indicates AWaRe tier. (b–d) Drugs comparatively defeated by *Klebsiella pneumoniae* on the same layout. Links accumulate near existing links, expanding from 12 drugs in 2014 to 21 in 2024 while distant regions remain dark. Highlighting is cumulative; degree normalisation affects only the layout. This is descriptive network expansion, not direct evidence of molecular acquisition or transmission.

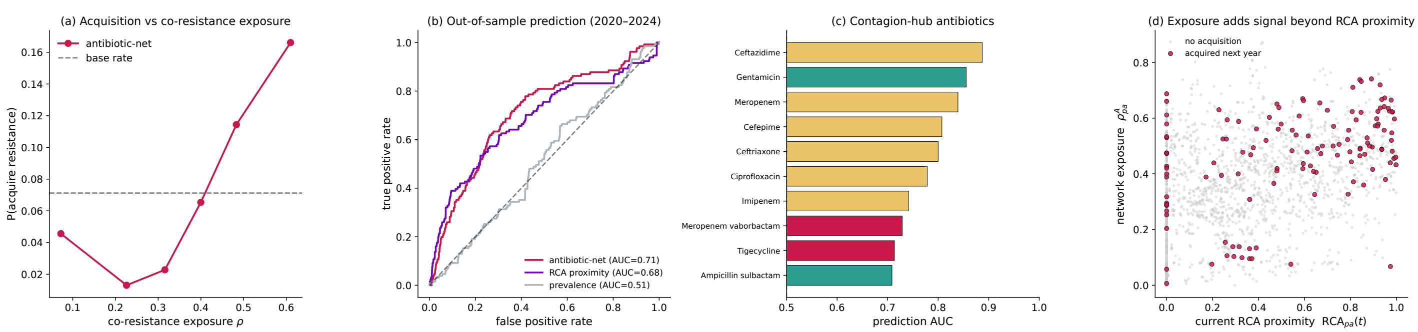

Figure 4: Forecasting the next drug. (a) One-year event frequency rises with network exposure. (b) Prospective ROC curves: network exposure (AUC 0.71), current RCA proximity (0.68) and drug prevalence (0.51). (c) Contagion hubs, the drugs whose neighbourhoods most reliably announce the next comparative-resistance event. (d) Exposure versus RCA proximity; future events can have high exposure even when current RCA is low. Prediction uses raw co-occurrence weights from M. The event is a new comparative-resistance link, not proof of biological acquisition.

References

[1] Jessica M. A. Blair, Mark A. Webber, Alison J. Baylay, David O. Ogbolu, and Laura J. V. Piddock. Molecular mechanisms of antibiotic resistance. *Nature Reviews Microbiology*, 13(1):42–51, 2015.

[2] Sally R. Partridge, Steven M. Kwong, Neville Firth, and Slade O. Jensen. Mobile genetic elements associated with antimicrobial resistance. *Clinical Microbiology Reviews*, 31(4):e00088–17, 2018.

[3] Pfizer. ATLAS Antimicrobial Surveillance Programme. Vivli AMR Register, 2024. Data coverage 2004–2024; `https://amr.vivli.org/`.

[4] Andrea Tacchella, Matthieu Cristelli, Guido Caldarelli, Andrea Gabrielli, and Luciano Pietronero. A new metrics for countries' fitness and products' complexity. *Scientific Reports*, 2:723, 2012.

[5] World Health Organization. AWaRe classification of antibiotics for evaluation and monitoring of use, 2023. Technical Report WHO/MHP/HPS/EML/2023.04, World Health Organization, 2023. `https://www.who.int/publications/i/item/WHO-MHP-HPS-EML-2023.04`.

[6] Evangelos I. Kritsotakis, Dimitra Lagoutari, Efstratios Michailellis, Ioannis Georgakakis, and Achilleas Gikas. Burden of multidrug and extensively drug-resistant ESKAPEE pathogens in a secondary hospital care setting in greece. *Epidemiology and Infection*, 150:e170, 2022.

[7] Giordano De Marzo, Alfredo De Bellis, and Chiara Mastrogiuseppe. Resistome atlas: Source code and reproducibility materials. `https://github.com/giordano-demarzo/resistome-atlas`, 2026. GitHub repository.

[8] Erdal Toprak, Adrian Veres, Jean-Baptiste Michel, Remy Chait, Daniel L. Hartl, and Roy Kishony. Evolutionary paths to antibiotic resistance under dynamically sustained drug selection. *Nature Genetics*, 44(1):101–105, 2012.

[9] Camilo Barbosa, Vincent Trebosc, Christian Kemmer, Philip Rosenstiel, Robert Beardmore, Hinrich Schulenburg, and Gunther Jansen. Alternative evolutionary paths to bacterial antibiotic resistance cause distinct collateral effects. *Molecular Biology and Evolution*, 34(9):2229–2244, 2017.

[10] César A. Hidalgo and Ricardo Hausmann. The building blocks of economic complexity. *Proceedings of the National Academy of Sciences of the United States of America*, 106(26):10570–10575, 2009.

[11] Jordi Bascompte, Pedro Jordano, Carlos J. Melián, and Jens M. Olesen. The nested assembly of plant–animal mutualistic networks. *Proceedings of the National Academy of Sciences of the United States of America*, 100(16):9383–9387, 2003.

[12] Manuel Sebastian Mariani, Zhuo-Ming Ren, Jordi Bascompte, and Claudio J. Tessone. Nestedness in complex networks: Observation, emergence, and implications. *Physics Reports*, 813:1–90, 2019.

[13] Hiroshi Nikaido. Molecular basis of bacterial outer membrane permeability revisited. *Microbiology and Molecular Biology Reviews*, 67(4):593–656, 2003.

[14] Roland Leclercq, Rafael Cantón, Derek F. J. Brown, Christian G. Giske, Peter Heisig, Alasdair P. MacGowan, Johan W. Mouton, Patrice Nordmann, Arne C. Rodloff, Gian Maria Rossolini, Claude-James Soussy, Martin Steinbakk, Tim G. Winstanley, and Gunnar Kahlmeter. EUCAST expert rules in antimicrobial susceptibility testing. *Clinical Microbiology and Infection*, 19(2):141–160, 2013.

[15] Inderjit S. Dhillon. Co-clustering documents and words using bipartite spectral graph partitioning. In *Proceedings of the Seventh ACM SIGKDD International Conference on Knowledge Discovery and Data Mining*, pages 269–274. Association for Computing Machinery, 2001.

[16] Albert Solé-Ribalta, Claudio J. Tessone, Manuel Sebastian Mariani, and Javier Borge-Holthoefer. Revealing in-block nestedness: Detection and benchmarking. *Physical Review E*, 97(6):062302, 2018.

[17] Daniele Laudati, Manuel Sebastian Mariani, Luciano Pietronero, and Andrea Zaccaria. The different structure of economic ecosystems at the scales of companies and countries. *Journal of Physics: Complexity*, 4(2):025011, 2023.

[18] Viktória Lázár, István Nagy, Róbert Spohn, et al. Genome-wide analysis captures the determinants of the antibiotic cross-resistance interaction network. *Nature Communications*, 5:4352, 2014.

[19] Matthew J. Ellington, Olov Ekelund, Frank M. Aarestrup, et al. The role of whole genome sequencing in antimicrobial susceptibility testing of bacteria: Report from the EUCAST subcommittee. *Clinical Microbiology and Infection*, 23(1):2–22, 2017.

[20] Elizabeth A. Ashley, Nandini Shetty, Jean Patel, Rogier van Doorn, Direk Limmathurotsakul, Nicholas A. Feasey, Iruka N. Okeke, and Sharon J. Peacock. Harnessing alternative sources of antimicrobial resistance data to support surveillance in low-resource settings. *Journal of Antimicrobial Chemotherapy*, 74(3):541–546, 2019.